

ZERO-SHOT SELF-ORCHESTRATION WITH LEDGER-BASED CONTROL IMPROVES CODING IN LANGUAGE MODELS

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ABSTRACT

Frontier coding performance is typically bought with larger proprietary models at high cost. We introduce ledger-based zero-shot self-orchestration, a training-free method in which fresh instances of one model decompose problems and coordinate through a shared filesystem holding a plan, notes and current solution. Across nine open and closed-weight models on the 100 latest *hard* LiveCodeBench problems, the method yields gains of up to 23.4 percentage points on pinned backends and offers two routes to near-frontier accuracy. Orchestrated GPT-5.6-Terra reaches 85.0% pass@1 against Fable 5’s 87.4% at 19% of the cost, and locally served, open-weight Qwen3.8-27B rises from 63.0% to 86.4%. Gains are not universal: some models are unchanged or worse. Transcript analysis attributes the gain to decomposition and persistent context. Inference-time organization can approach frontier coding accuracy at a fraction of the cost, or on self-hostable weights.

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed not as single-shot predictors but as *agents*: models wrapped in a scaffold that lets them plan, call tools, keep notes, and revise their work over multiple turns. A popular approach to make LLMs is to run *several* such agents together, on the intuition that a team that decomposes a problem and pools partial results should beat a single model answering in one pass. That intuition has produced a large literature (Guo et al., 2024), but the evidence is mixed (Wang et al., 2024b) and much of it confounded (Tran & Kiela, 2026): multi-agent pipelines usually change token budgets, tool calls, prompts and retrieval at once, so an aggregate gain rarely identifies which factor helped.

We use *agent* to mean a separately prompted model invocation with a distinct role and a fresh context; all agents share one underlying model. We call the method studied here zero-shot self-orchestration: the orchestrator is neither trained for orchestration nor given task-specific demonstrations of how to decompose or coordinate the problem. Three properties define it: a shared filesystem workspace — the *ledger* — that persists a plan, notes and current solution across invocations; a manager that re-curates the task list each round and decides when to stop, rather than following a fixed pipeline; and no training or per-benchmark tuning, every model role being the same model in a fresh context with a short generic prompt. Section 3 describes the loop and Appendix A gives it in full.

On the 100 most recent *hard* LiveCodeBench problems (Jain et al., 2024), our training-free filesystem scaffold improves all three models tested in both arms on a pinned backend by 8.0 to 23.4 points across five paired passes, despite using no router, reward model, search policy, or benchmark-specific tuning (Sections 3 and 5.1). Although the scaffold roughly triples the bill, orchestrating a cheaper model remains the less expensive route to comparable accuracy in every supported comparison: orchestrated GPT-5.6-Terra trails Claude Fable 5 by 2.4 points while costing \$11.71 rather than \$61.11 per pass, Qwen3.8-27B comes within 1.0 point, and GPT-5.6-Luna matches GPT-5.6-Terra’s single-call performance at 44% of the price (Section 5.2). These gains diminish as single-call performance rises and can disappear or reverse: two of the nine models come out flat or negative, with Qwen3.6-35B’s transcripts under reasoning off showing deliberation discarding correct approaches and unverified worker output overwriting correct work (Sections 5.4 and 6). We also correct a

LiveCodeBench evaluator bug that penalizes a common input-reading idiom and may therefore affect previously reported scores (Appendix C).

2 RELATED WORK

Multi-agent systems build on single-agent scaffolds that add structure around one model (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023); our worker role is an instance of that idea, and what we vary is the coordination layer above it. Prior systems differ mainly in how that layer is decided. One line *trains* the coordinator: Fugu (Sakana AI, 2026) learns to assemble and direct a pool of expert workers per query, and Conductor (Nielsen et al., 2026) is a 7B model RL-trained to design worker communication topologies; earlier query routers learn to pick a model per query (Ong et al., 2025). A second fixes roles and hand-off order in advance, so every problem flows through the same pipeline (Hong et al., 2024; Qian et al., 2024; Li et al., 2023; Huang et al., 2023; Islam et al., 2024), while frameworks such as AutoGen (Wu et al., 2024) instead supply configurable conversable agents and leave the conversation pattern open; Trinity (Xu et al., 2026) sits at the boundary with the first, fixing a three-role workflow but learning which model fills each role. A third coordinates through a shared store rather than a pipeline — ARIADNE (Wei et al., 2026) pairs a blackboard with MCTS on program generation, LbMAS (Han & Zhang, 2025) mediates all communication through a global board — or decentralizes entirely, through debate (Du et al., 2024) or sampling and voting (Li et al., 2024); Mixture-of-Agents (Wang et al., 2025) instead aggregates in layers. A separate line asks not how coordination is decided but whether reported gains survive equalizing test-time compute (Wang et al., 2024a; Tran & Kiela, 2026).

Ours is the training-free counterpart to the first family and the dynamic counterpart to the second. The manager is a fixed prompt with no learning, which lets us ask how much of the orchestration benefit is available *without* training, and when it materializes. Its workspace is a shared store like ARIADNE’s blackboard, but coordination is one manager’s plain task-list loop: no MCTS, no reward model, no learned search policy. We do not attempt an equal-token comparison — the loop necessarily spends more than a single call — and instead ask whether that spend buys better solutions, and whether it buys them more cheaply than moving to a larger model would, measured throughout against the same model answering in one call.

3 METHOD

Every role but the verifier is the same underlying model, called in a fresh context and coordinating through a shared workspace on disk:

<code><ws>/task.md</code>	the problem statement
<code><ws>/plan.md</code>	the manager’s overarching plan
<code><ws>/tasks.json</code>	a debug mirror of the task list
<code><ws>/notes.md</code>	curated ideas / findings / partial proofs
<code><ws>/solution.py</code>	the current solution

Workers never talk to each other; each sees the ledger plus the one task the manager assigns. The control flow (Figure 1) is:

1. **Manager — plan.** The manager reads the problem and writes a 3–6 sentence strategy plus 3–6 concrete seed tasks.
2. **Worker — brainstorm (ideation).** The first worker does *not* write a solution; it identifies the core difficulty, lists candidate approaches and pitfalls, and appends them to `notes.md`, proposing next steps.
3. **Manager — manage (loop).** The manager folds the plan and the brainstorm into one curated task list (merging duplicates, marking done items, adding only genuinely new sub-tasks), then either declares the problem done or names the single next task.
4. **Worker — do the task.** A fresh worker executes that one task and rewrites both `solution.py` and `notes.md` — the notes are replaced by a curated whole rather than appended to — then proposes remaining steps.

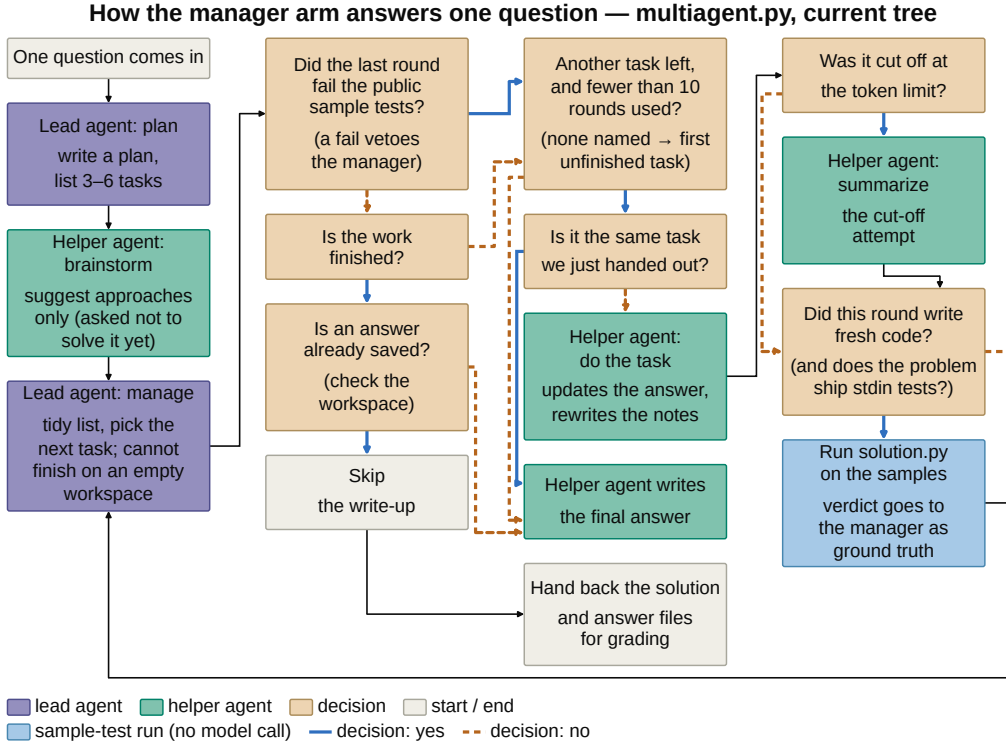


Figure 1: **Manager–worker control flow**, as the v2 scaffold runs it. Boxes labelled with an agent are model calls in a fresh context; the decisions, the start and end nodes, and the sample-test run are harness logic and call no model. The boxes absent from the original scaffold are the verifier of step 5 and the cut-off summarizer.

5. **Verifier — run the public tests.** Whenever the round’s worker produced a fresh candidate, the program is executed against the problem’s public stdin tests where it has them. The pass/fail verdict, with the first failing case, is fed back to the manager and treated as ground truth: a failing run overrides a done verdict, forcing the loop to continue with a fix-or-switch task. Control returns to the manager (step 3).
6. **Finalizer.** A finalization worker emits the definitive solution whenever the loop ends without a clean sign-off — the round budget is spent, the manager reissues a task it just handed out, or it names no task at all. The call is skipped when the manager declares the problem done and a usable solution is already on disk, so a redundant final pass cannot overwrite a correct answer.

Guards: a round budget of 10 manager→worker cycles (`MAX_ITERS`); a no-progress guard (if the manager reissues the exact task it just handed out, the loop stops); and a cut-off summarizer: a worker that hits the token cap mid-attempt has its partial reasoning summarized by a fresh short call so its ideas still reach the manager. The plan and notes files are capped when written, at 4,000 and 8,000 characters, so neither grows without bound across rounds.

Baseline. The single-call baseline is the same model, at the same temperature as the manager arm’s workers (0.2), given the same problem in exactly one call — no shared workspace, no loop, and no other role in the prompt. The single call receives the solver system prompt verbatim; the manager arm’s task workers receive that same prompt wrapped in a subagent preamble and a four-section output contract, and their user message additionally carries the plan, the current notes and the current solution. The two arms differ only in that scaffold. Appendix A gives the per-role temperatures and the five differences between this scaffold (v2) and the original one that produced the OpenRouter-served results of Section 5.4.

4 EXPERIMENTAL SETUP

Benchmark. LiveCodeBench code-generation, release_v6 (Jain et al., 2024): the 100 most recent problems in the hard split by contest date, which limits training-data contamination (Jain et al., 2024). We refer to this set as LCB-100. Solutions are graded by LiveCodeBench’s own hidden-test evaluator, after the correction described in Appendix C; every number below is post-correction. The verifier of step 5 runs only stdin-format public tests, which 73 of the 100 problems carry.

Models and serving. Nine model configurations: Qwen3.5-9B (Qwen Team, 2026a), Qwen3.6-35B-A3B (Qwen Team, 2026b), Qwen3.8-27B (Qwen Team, 2026c) (open weights, served locally in FP8), MiniMax-M3 (MiniMax, 2026), Kimi-K3 (Moonshot AI, 2026), and the closed frontier models Opus-5 (Anthropic, 2026a), the two GPT-5.6 variants Terra and Luna (OpenAI, 2026a), and Claude Fable 5 (Anthropic, 2026a). Serving splits the set in two. The four *pinned-backend* models are served on one fixed backend for v2 comparison: Terra and Luna by the OpenAI API, Qwen3.8-27B by our own vLLM, Fable 5 by the Anthropic Messages API; because gateway routing proved the dominant source of run-to-run noise (Section 6). The five *OpenRouter-served* models ran through a single OpenAI-compatible gateway on the original scaffold, and are reported separately in Section 5.4 because both the serving path and the scaffold version differ. Fable 5 was run single-only, so it contributes no manager delta.

Conditions. The pinned-backend set runs at one setting — 128k output cap, reasoning on, $\times 5$ passes — with Terra, Luna and Qwen3.8-27B in both arms. The cap is a per-call `max_tokens` bound, not a context-window limit. The OpenRouter-served set runs at three: 128k reasoning-on $\times 1$; 16k reasoning-off $\times 5$; and 128k reasoning-off $\times 1$ as a control isolating the small cap. Appendix B gives the reasoning-effort settings per provider, the instrumentation, and the cap-matching replay used for Qwen3.8-27B’s single arm.

Statistics. Reported $\pm$ SD is the sample SD (divisor $n - 1 = 4$) of the five pass scores. Five-pass comparisons use a paired sign-flip permutation test on per-problem mean Δ ($n = 100$ problems; 2×10^5 resamples), Holm-corrected where a family of models is tested; 95% t -intervals for mean pass@1 across the five passes (df = 4). Pooled problem $\times$ pass agreement uses exact McNemar on the discordants, treating cells as independent.

5 RESULTS

5.1 PINNED-BACKEND ARMS, FIVE PASSES

Our headline condition is four models, each run five independent times on LCB-100 at a 128k cap with reasoning on. Every arm is served on a pinned backend and runs the v2 scaffold except Fable 5. The earlier runs of the OpenRouter-served set (Section 5.4) cover more models, but only their 16k reasoning-off condition repeats; the rest are one pass each, through a gateway whose routing proved the main source of run-to-run noise.

All three models with both arms benefit from the manager. It also narrows run-to-run spread wherever that spread was wide, for Luna and Qwen3.8-27B alike. Luna’s per-pass Δ swings across a wide band, so its effect is real but its per-pass estimate unstable; Terra’s flat band is the more reliable. Qwen3.8-27B gains most, its manager arm reaching Fable 5’s single-call range here, and Opus-5’s in the OpenRouter-served set. The direction matches that set’s reasoning-on pattern: the smaller the model, the more the scaffold has to add.

Figure 2 orders the four models by single-call score: the manager’s gain shrinks monotonically as the single call gets stronger, with two of the manager arms here coming within a few points of Fable 5’s single call, whose lone bar marks where that trend is heading.

5.2 WHAT THE SCAFFOLD COSTS

Every Δ in the pinned-backend set is bought with tokens. The manager replaces one call with a plan, a brainstorm, up to ten worker rounds and a finalization pass, and every one of those is billed. The

Manager vs single call, four models —
LCB-100, 5 passes, 128k max tokens, reasoning ON

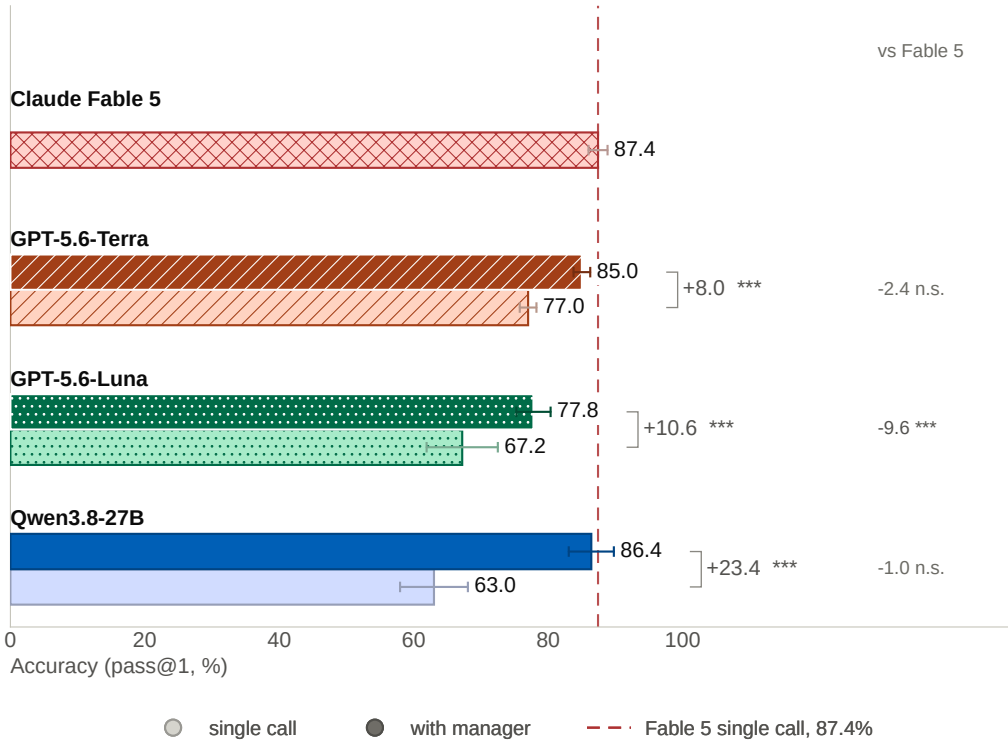


Figure 2: **Manager vs. single call.** 128k $\times$ 5 passes, reasoning on. Bars are pass@1 on the same 100 problems, the line through each the 95% CI across the 5 passes (t, df = 4). The top bar of each pair is the manager, dark; the single call is under it, light; hatch is the model. “Single call” is one call with no tools and no loop; Fable 5 ran single-only, so it has one bar, which the dashed rule carries across the chart. The bracket beside each pair is manager – single call; the right-hand column tests the top bar against Fable 5. Both are paired sign-flip permutation tests, unit = problem (n = 100), Holm-corrected within each family of 3: * p < .05, ** p < .01, *** p < .001. The three within-model Δ all clear p < 1e-4; against Fable 5, p = GPT-5.6-Terra 0.59, GPT-5.6-Luna 4.6×10^{-4} , Qwen3.8-27B 0.73. Manager-only vs single-only problem-passes (500 per model): GPT-5.6-Terra 51/11; GPT-5.6-Luna 71/18; Qwen3.8-27B 125/8; exact McNemar 3×10^{-7} or better. The gains are not compensating wins and losses. Every arm is at a 128k cap and re-scored on the corrected evaluator (Appendix C); Qwen3.8-27B’s single arm is the 128k cap-matched replay of a 250k generation, which Appendix B describes and Section 5.2 prices.

scaffold roughly triples the bill. Every arm priced here is at the same 128k output cap, Qwen3.8-27B’s single call included (Appendix B).

The scaffold’s move is up and to the right in every case. Figure 4 plots price against accuracy for all seven arms, joining the two arms of each model that has both: every manager arm gains accuracy over its own single call, and every one costs more. What the extra dollar buys varies by an order of magnitude, tracking the spread in base prices. The frontier is nearly flat at the top.

The base cost varies across models. A single-call pass on Qwen3.8-27B emits more than twenty times the output tokens of either OpenAI model: the open model reasons its way to its score at length; the OpenAI models reach theirs in a fraction of the tokens. Since reasoning bills as output, a bill here is as much a fact about how much reasoning a provider lets a model emit as about the scaffold wrapped around it.

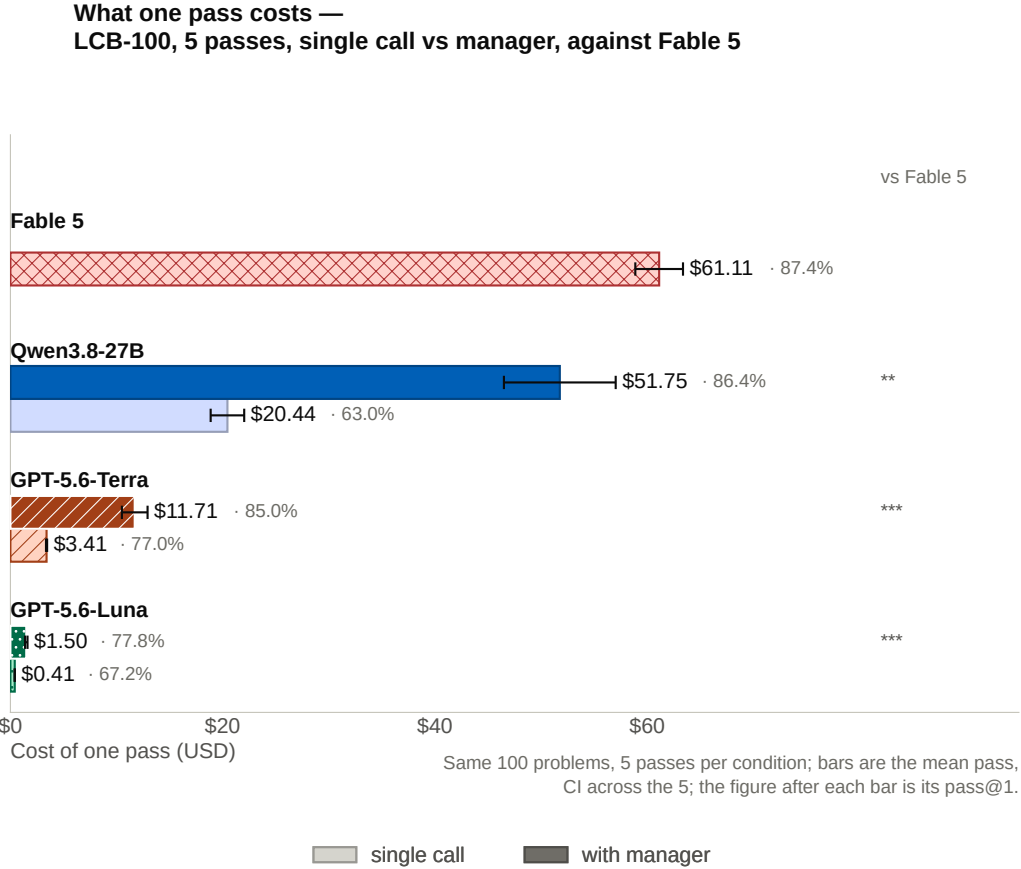


Figure 3: **The scaffold’s bill.** Cost of one pass over the same 100 problems; Table 1 is the arithmetic behind every bar. No cached-input discount is taken, and Qwen3.8-27B is priced at OpenRouter market rates. The top bar of each pair is the manager, dark; the single call is under it, light; all are at a 128k output cap, and hatch is the model. The right-hand column tests each top bar against Fable 5’s single call (Welch per run, $n = 5$ vs 5; Holm-corrected across seven comparisons): * $p < .05$, ** $p < .01$, *** $p < .001$. Unbracketed comparisons: each manager – single increase is significant (8.7×10^{-5} or better per run, $< 3.5 \times 10^{-5}$ per problem); GPT-5.6-Luna’s manager undercuts GPT-5.6-Terra’s single call by \$1.91 (2.1×10^{-7} per run, $< 3.5 \times 10^{-5}$ per problem). At 1.18x the assumed Qwen rate (\$0.413/\$3.25 per MTok, inside the spread across hosted providers) the manager’s cost advantage over Fable 5 disappears entirely — a larger uncertainty than any p-value here.

Three comparisons along the frontier. First, GPT-5.6-Terra with a manager nearly matches Fable 5 for a fifth of the price (Figure 3). The gap is within error (Figure 2), with a one-sided 95% bound allowing a deficit of 5.8 points, while the saving is significant. Both arms are priced off API list rates in effect when the experiments were run (OpenAI, 2026b; Anthropic, 2026c).

Second, Qwen3.8-27B with a manager comes within a point of Fable 5’s accuracy and costs less. That difference is again within error, the one-sided 95% bound allowing a deficit of up to 4.8 points. The cost uses a third-party OpenRouter host rate (OpenRouter, 2026); our test ran locally, and anyone with capable hardware could host it for less.

Third, the two GPT-5.6 models make the clearest case for spending on the scaffold rather than on the model: GPT-5.6-Luna with a manager matches GPT-5.6-Terra’s single call on accuracy at under half the price. Luna’s lead is not itself significant, but there is no sign of a deficit either, and the bound rules out its trailing by more than 2.4 points.

5.3 THE EFFECT OF TRUNCATION

Empty solutions are automatic fails, so any condition that emits fewer of them gains pass@1 without solving more problems. Per-call `finish_reason` separates that channel from genuine problem-solving exactly (Table 2). Neither GPT-5.6 model truncates or emits an empty answer across all 2,000 of their problem-passes, so their Δ s in the pinned-backend set carry no rescue component. The channel is largest on Qwen3.8-27B, and even there a cut-off rate is not a loss rate: most truncated generations still contain a complete solution, written inside the reasoning stream before the cap arrived and recovered from there by the extractor. Of the single arm’s 35 empty cells, the manager solves 25, fails 10 and leaves none empty, worth 5.0 points, about a fifth of its gain — a cut-off worker costs a round rather than the answer. Served locally at a 262,144-token context, Qwen3.8-27B’s single arm was in fact generated at 250k and cap-matched to 128k afterwards (Appendix B); read at its original 250k cap it gains a little more, by shrinking the truncation tail rather than lifting the run (Table 3). That reading is exploratory, cap-matched to nothing else in the paper, so it cannot carry a Δ against the manager arm. Re-scoring each single arm over only the problem-passes where it emitted code bounds the effect model by model (Table 4); Table 5 gains 1.6 points there, six of its nine empty cells being safety refusals rather than truncations (Section 6).

5.4 THE OPENROUTER-SERVED MODEL SET

The five models below were served through the OpenRouter gateway and run on the original scaffold, so they are not directly comparable to the pinned-backend arms above. Table 5 collects every condition.

Reasoning on. Opus-5 tops the study, its +6 a net of seven problems gained and one lost. The largest gains come from the smaller models (Figure 5a): ordered by scale, the single curve rises further than the manager curve, so the scaffold narrows the spread across model sizes (Figure 6, which plots the same numbers against parameter count).

Reasoning off. We ran five independent passes per condition at a 16k cap plus a one-pass 128k control. Run-to-run spread across the five passes is small. The manager effect is significant for three of four models, and a Tukey HSD separates Kimi-K3 from all three others and MiniMax-M3 from Qwen3.6-35B; Qwen3.5-9B sits between those two and separates from neither (Figure 5b; Figure 7 by scale).

The 128k control largely eliminates truncation and reproduces the ordering (Figure 5c). Empty output is not fully removed — single-arm emit rates (non-empty completions) run below the manager’s throughout, so part of the gap at the small end remains an emit-rate effect — but it cannot explain Kimi-K3, where a 4-point emit gap accompanies a score gap ten times larger and neither of its arms truncates here at all. Unlike the reasoning-on setting, the single and manager curves diverge here (Figure 8), though the two settings span different model sets.

6 DISCUSSION

Table 6 collects one transcript per model in which the single call failed and the manager arm solved the problem. Three mechanisms recur. The scaffold commits the reduction or the bounding argument to disk before any code is written, so the next call builds on it instead of re-deriving it. It splits coupled objectives into pieces that are implemented separately. And it bounds each call, so a model that would otherwise talk past its output cap gets its work written down. In two cases the manager’s solution is also markedly smaller than the failed single-call one.

Truncation is a second place the scaffold earns its keep. State lives in `plan.md`, `notes.md` and `solution.py` rather than in one ever-growing transcript, and each worker sees a curated view of it — plan, notes, current solution, one task — instead of the full history. The workspace is external memory that the token cap cannot truncate.

That matters because a generation left to run to a very large cap does not merely risk being cut off; it can degrade the model. Eight of the fourteen Qwen3.8-27B single calls that spent a full 250k budget without emitting code collapse into repetition — on `abc399_e`, 7,743 copies of one line — most of

the eight inventing edge cases against a candidate solution and answering each in turn, hundreds to thousands of times, without terminating. This is the mechanism behind the otherwise odd finding of Section 5.3 that most cut-off generations still contain a complete solution: the model had reached an answer and then talked past the cap re-checking it. Bounding each call keeps the work far from the cap and gets it onto disk.

The same argument should extend beyond this benchmark, though we do not test it. A task whose inputs or intended output do not fit in one call — a codebase larger than the context window, a deliverable longer than the output ceiling — is not solvable in one call at any reasoning effort, whereas a sequence of bounded steps can carry it.

The manager helps most where the base model is least able to organize its own reasoning, and that weakness takes two forms. The first is having no internal planning stage at all: with reasoning disabled a single call jumps straight to code, and the scaffold supplies a planning space, on disk and across calls, that the model would not otherwise use (Figure 5c). The second is reasoning that runs away: Qwen3.6-35B with reasoning on spends its whole budget deliberating, and gains more here than in any of its other conditions (Figure 5a). Where a model can both plan internally and stop on its own, the scaffold has less to add.

The manager is not free: its deliberation can settle on a worse answer than a single call would have produced. On LCB 3765 (Qwen3.6-35B, 128k, reasoning off) the ideation stage identified the same optimization the single call had used, talked itself out of it as too error-prone, and committed instead to an asymptotically slower plan that a worker then implemented with a further bug — compact, correct code disturbed rather than helped by decomposition. Qwen3.6-35B with reasoning off is the standout loser on this pattern, negative at both caps.

Two inexpensive, training-free additions could address the two halves of that failure, the plan that was wrong and the implementation that went unchecked. First, *fresh-perspective workers*. Every worker inherits the notes and the current solution, so a wrong early approach anchors everything downstream. Spawning some workers with the raw problem statement and no prior context would give the manager an independent attempt to compare against, protecting the tight single-call solution the scaffold otherwise disturbs — the intuition behind sampling-and-voting and debate (Li et al., 2024; Du et al., 2024). Second, *verification instead of trust*: the manager currently accepts a worker’s report on the basis of a sanity check against the problem’s own samples, so a confidently-wrong solution propagates unchecked and can overwrite a correct intermediate answer. Running each candidate against a generated test suite before acceptance would catch regressions of this kind, at the cost of extra calls.

6.1 LIMITATIONS

- **Serving-provider reliability (OpenRouter-served set only).** The five models of that set ran through OpenRouter’s multi-provider routing, a substantial and *time-varying* confound: providers intermittently stalled, dropped the response mid-stream, returned error objects, clamped output below the requested cap, or returned reasoning-only replies. An attempt with no usable completion is scored a failure, so its pass@1 counts infrastructure failures as wrong answers — a conservative lower bound, and not exactly reproducible, since it depends on provider health at run time. Both arms draw on the same provider pool, so the single-vs-manager comparison is less sensitive to this than the absolute levels are. This confound is what motivated the pinned-backend set, which is why those arms and not the OpenRouter sweep carry the paper’s headline claims.
- **Provider-served weights may vary (OpenRouter-served set only).** OpenRouter does not guarantee one quantization or build per model, adding a second, smaller source of run-to-run variation on top of decoding temperature. The headline arms are pinned to one backend each, and the locally served Qwen3.8-27B to a specific FP8 checkpoint.
- **Single pass in the OpenRouter-served set.** The reasoning-on and 128k reasoning-off comparisons there are one pass per condition, to limit API cost, so their deltas are point estimates within the ~ 4.5 percentage points of pass-to-pass variation the 16k five-pass runs show on a manager-minus-single delta; per-model significance rests on those five-pass runs. It is also why Opus-5’s 85 $\rightarrow$ 91, the highest score in the paper, remains a point estimate rather than a measured effect.

- **Table 5 has no manager arm.** It was run single-only, so the strongest single-call result in the paper is also the one condition where we cannot say what the scaffold would do.
- **Refusals are scored as failures, and only one arm can incur them.** Table 5 returned `stop_reason=refusal` on 6 of 500 problem-passes, spread over three ordinary competitive-programming problems and never deterministic. We deliberately configure no fallback model, since a cell labeled Table 5 must contain Table 5’s outcome, but the consequence is that its score carries ~ 1.2 points of classifier penalty no other arm is exposed to, as variance rather than a constant offset. Comparisons *against* Table 5 are therefore mildly conservative in its disfavor.
- **Qwen3.8-27B’s 128k single arm is a replay, not an independent run.** Those generations were produced at 250k and truncated to 128k after the fact (Appendix B). It is the right comparison for isolating the cap — identical generations either side — but it cannot capture how the model might have budgeted its reasoning had it been told the smaller limit up front; we report the as-generated 250k reading alongside it (Section 5.3).
- **A 9B model with reasoning on was untestable.** Qwen3.5-9B with reasoning enabled could not be evaluated in any configuration we tried: through OpenRouter it generates unbounded reasoning and returns reasoning-only replies or truncates before an answer appears, and no token or effort setting fixed it; locally the checkpoint refused batching, leaving zero completed calls. We report it as a model limitation rather than a data point, and it bounds how small a “reasoning” model this scaffold can be applied to.
- **One benchmark family.** All reported results are competitive-programming code generation. Math and knowledge benchmarks (AIME, MATH-500, GPQA, HLE) were run in exploratory form but are not reported: on a probe of the ten hardest problems in each, the frontier models sit at or near ceiling, leaving little headroom in which to measure a scaffold effect.

7 CONCLUSION

With the underlying model held fixed, a training-free manager–worker scaffold over a shared filesystem improves coding accuracy on every pinned-backend model we ran in both arms. Nothing in the loop is learned: the coordination is a directory of files that successive invocations of the same model read and overwrite, driven by one short generic prompt. Transcript analysis attributes the gain to decomposition and to context that persists across calls.

The scaffold roughly triples the token bill, yet in every comparison our data supports, orchestrating a cheaper model was the less expensive route to a given accuracy than switching to a larger one. That leaves two ways to approach frontier coding accuracy without paying frontier prices: orchestrate a cheaper API model, or orchestrate open weights small enough to serve locally. The result is conditional rather than general — the gains shrink as the single call gets stronger, and some configurations come out flat or worse, where deliberation displaces an approach that was already right — but where it holds, inference-time organization substitutes for model scale.

AI USE STATEMENT

In this work, we used generative AI tools as the object of study: every experimental result is a measurement of language-model output, produced by the scaffold and baselines described in Section 3 and released with the code. Separately, generative AI tools assisted with writing and editing the manuscript, with implementing the orchestration harness and the analysis and plotting scripts, and with literature search. We did not use generative AI tools to generate, impute or alter any experimental result, and all reported numbers are computed by the released scripts directly from the stored run outputs. We have reviewed all AI-assisted work: the harness and analysis code was read and tested by the authors, the corrected evaluator of Appendix C was found and verified by manual inspection of transcripts, and every citation was checked against its primary source. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work involves no human subjects, no personal data and no new dataset release; the evaluation set is the public *hard* split of LiveCodeBench release_v6 (Jain et al., 2024). Two points bear on research integrity. First, one arm returned safety refusals that our protocol scores as failures (Section 6.1); we report the rate rather than substituting a fallback model, so a cell labeled with a model contains that model’s outcome. Second, we report a defect in a widely used public evaluation harness (Appendix C) and the corrected re-scoring of every number, rather than only the numbers that favor our method. The method itself is a general-purpose inference-time scaffold for code generation and inherits the dual-use profile of the underlying models; it lowers the cost of capable code generation, which applies to beneficial and to harmful code alike.

REPRODUCIBILITY STATEMENT

Section 3 and Appendix A specify the manager–worker loop, its guards and its per-role temperatures; Section 4 and Appendix B give the benchmark selection, the serving path and provider settings per model, the output caps and pass counts, the instrumentation, and the cap-matching replay procedure for Qwen3.8-27B’s single arm. Appendix C documents the evaluator defect and the fix, without which the reported numbers cannot be reproduced from the stored generations. The statistical procedures are given in Section 4. Two sources of irreducibility are stated in Section 6.1: decoding temperature is non-zero, and the OpenRouter-served arms depend on provider health at run time. The pinned-backend arms, which carry the paper’s claims, avoid the second. The scaffold, the corrected evaluator, the analysis and figure scripts, and all run outputs supporting the results — per-problem generations for every arm and condition, per-call `finish_reason` and token logs where instrumented, and the manager arms’ workspace transcripts — are provided as anonymized supplementary material, and will be released publicly on acceptance.

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A THE SCAFFOLD IN FULL

Section 3 describes the current scaffold, v2, which produced the pinned-backend results and is drawn in Figure 1. The original scaffold produced the OpenRouter-served set of Section 5.4. It runs the same loop on closely matched prompts, with five differences, all of them on the manager arm: it allows 4 manager→worker cycles rather than 10 (`MAX_ITSERS`); it has no public-test verifier at step 5; it has no cut-off summarizer; its workers append to `notes.md` rather than rewriting it; and the plan and notes files feeding a prompt are unbounded rather than capped.

Temperatures. The manager arm’s generative calls use slightly higher temperatures, where the work is generative rather than executive: 0.3 to write the plan and 0.4 to brainstorm, against 0.2 for task execution and for curating the task list. The single-call baseline runs at 0.2, matching the workers. These settings are identical in both scaffold versions.

Verifier coverage. The step-5 verifier runs only stdin-format public tests, which 73 of the 100 problems carry; the other 27 are LeetCode-style, with call-based public tests the engine does not run, so those get no verifier signal and are checked by the hidden-test grader alone.

B BENCHMARK AND PROTOCOL IN FULL

Model sizes. Qwen3.5-9B (Qwen Team, 2026a) (9B), Qwen3.6-35B-A3B (Qwen Team, 2026b) (35B total, 3B active per token), Qwen3.8-27B (Qwen Team, 2026c) (27B, open weights, served locally in FP8), MiniMax-M3 (MiniMax, 2026) (428B total, ~23B active), Kimi-K3 (Moonshot AI, 2026) (~2.8T total, ~16 of 896 experts active per token), and the closed frontier models Opus-5 (Anthropic, 2026a), GPT-5.6-Terra and GPT-5.6-Luna (OpenAI, 2026a) and Claude Fable 5 (Anthropic, 2026a), all four of undisclosed size.

Caps. The 128k figure is a per-call `max_tokens` bound, not a context-window limit; for Opus-5, Fable 5 and the GPT-5.6 family it is also the provider’s hard ceiling on one response (Anthropic, 2026a; OpenAI, 2026a). Locally served Qwen3.8-27B has no such ceiling; it is bounded by its 262,144-token context window (Qwen Team, 2026c) — confirmed on the FP8 checkpoint as served (vLLM reports `max_model_len = 262,144`) — shared by prompt and output, which puts the 250k setting of Section 5.3 near its largest possible output.

Conditions. The pinned-backend set runs at one setting: 128k output cap, reasoning on, $\times 5$ passes, with Terra, Luna and Qwen3.8-27B in both arms and Fable 5 single-only. The OpenRouter-served set runs at three: 128k reasoning-on $\times 1$ (native `reasoning_effort` where supported, a 20k reasoning budget where not); 16k reasoning-off $\times 5$, whose repeated passes give run-to-run variance and paired tests; and 128k reasoning-off $\times 1$, the same without the small cap, isolating it as a confound.

Reasoning. The output cap and the scaffold are matched across the seven pinned-backend arms; reasoning depth is not, because no two of these providers expose the same control over it. Qwen3.8-27B, on our own vLLM, is sent no `reasoning_effort` — the Qwen chat template leaves reasoning on and unbudgeted, bounded only by the 128k `max_tokens`, which on this stack covers reasoning

and answer together. The two GPT-5.6 models are likewise sent none, leaving them at OpenAI’s model default. Fable 5 runs at `thinking: adaptive with display: summarized`, and `output_config.effort` at high, the API default (Anthropic, 2026b).

Instrumentation. The two 128k-cap conditions log every call’s `finish_reason` and completion-token count, and give each problem a final status in `{ok, truncated, empty_stop, empty, error}`. The $16k \times 5$ -pass runs predate this and record only the extracted code and its pass/fail. An empty code field scores as a failure.

Cap-matching. One arm is not natively cap-matched to its partner: Qwen3.8-27B’s single call was generated at 250k while its manager arm ran at 128k (Section 5.3). To compare them at one cap we replay the stored generations rather than re-run the model: each is truncated to 128,000 output tokens, the solution re-extracted from that prefix, and the result scored on the same evaluator as every other cell (Appendix C). Three details matter:

- **Truncation is token-exact**, using the serving stack’s own tokenizer, not a character-count approximation. The deciding cases are those where a complete code block sits just before or just after the boundary — exactly the ones an approximate cut would misclassify.
- **What is truncated is the whole output stream, reasoning then answer**, because that is what the original cap bounded. Where the cut lands decides what the extractor sees: past the reasoning it sees a truncated answer; inside the reasoning the answer does not exist at all, and the harness’s empty-content fallback hands it the truncated reasoning instead — which is why so many cut-off generations still yield gradeable code (Section 5.3).
- **The cap is invisible to the model, which is what makes the replay exact.** This arm ran on our own vLLM, where `max_tokens` is enforced by the server: it never enters the prompt, and generation simply cuts off at the limit — 124 of the 500 calls end at `finish_reason=length`, abruptly rather than wound up. Nor is this local to vLLM: making a budget visible to a model takes a separate, opt-in mechanism such as Anthropic’s `task_budget`, while `max_tokens` stays an enforced ceiling that truncates the response (Anthropic, 2026b). The replay therefore does not differ from a genuine re-run at 128k.

C A CORRECTION TO THE LIVECODEBENCH EVALUATOR

While inspecting transcripts for Section 6 we found a defect in the LiveCodeBench harness itself, and every number in Section 5 is reported after fixing it. The evaluator does not run a candidate as a subprocess; it executes it in-process with `sys.stdin` replaced by a mock. That mock’s binary view implemented `readline()` as `inputs.split(b"\n")[0]` — a *stateless* expression that returns the first line on every call. A program reading multi-line input through `sys.stdin.buffer.readline()` therefore read line 1 repeatedly and failed however correct it was. The mock’s text-mode `sys.stdin.readline` was implemented as a proper iterator, and `buffer.read()` returns the whole payload, so neither was affected — which is why the common `sys.stdin.buffer.read().split()` idiom never exposed it. We replaced the mock’s binary view with one backed by a `BytesIO` so that reads advance a position, and re-scored every stored generation; no model was re-run.

We report it explicitly for two reasons. First, it interacts badly with the v2 verifier: that step runs the candidate as a *real* subprocess (step 5 of Section 3), where `buffer.readline()` works correctly, so the manager was told its solution passed the public samples for programs the grader then marked wrong — the loop’s one external signal certifying a wrong answer. Second, the exposure is uneven across models rather than a constant offset: 311 of the 3,456 pinned-backend outputs containing code use the idiom, and 97% of those were scored wrong against 20% for the outputs that do not. Fable 5 never uses it and its scores are unchanged to the decimal. The OpenRouter-served set reaches for it once in 5,012 generations containing code, and then through an alias — Kimi-K3’s single call on `abc396_g` binds `sys.stdin.buffer` to a name and calls `readline()` on that. Every OpenRouter-served number is re-scored all the same, and exactly one cell moves: Kimi-K3’s 128k reasoning-on single arm, from 82 to 83. Because the idiom is a matter of coding style, a harness bug of this shape is not a wash across a leaderboard — it silently penalizes the models whose style trips it.

D ADDITIONAL RESULTS

Table 1: **What one pass cost each arm.** List rate $\times$ the tokens it consumed.

Arm	Rate \$/MTok in / out	In (MTok)	Out (MTok)	\$/pass	\$/solved
Qwen3.8-27B single	\$0.35 / \$2.75	0.0753	7.4247	\$20.44	\$0.32
Qwen3.8-27B manager	\$0.35 / \$2.75	1.5053	18.6277	\$51.75	\$0.60
GPT-5.6-Luna single	\$0.20 / \$1.20	0.0661	0.3299	\$0.41	\$0.006
GPT-5.6-Luna manager	\$0.20 / \$1.20	1.1686	1.0522	\$1.50	\$0.019
GPT-5.6-Terra single	\$2 / \$12	0.0661	0.2728	\$3.41	\$0.044
GPT-5.6-Terra manager	\$2 / \$12	1.1098	0.7911	\$11.71	\$0.14
Fable 5 single	\$10 / \$50	0.0899	1.2043	\$61.11	\$0.70

List rates: Qwen3.8-27B (OpenRouter, 2026), GPT-5.6-Luna and GPT-5.6-Terra (OpenAI, 2026b), Fable 5 (Anthropic, 2026c). OpenAI rates are the short-context tier and Anthropic’s the base rates; no long-context tier, Batch discount or prompt-caching multiplier is applied.

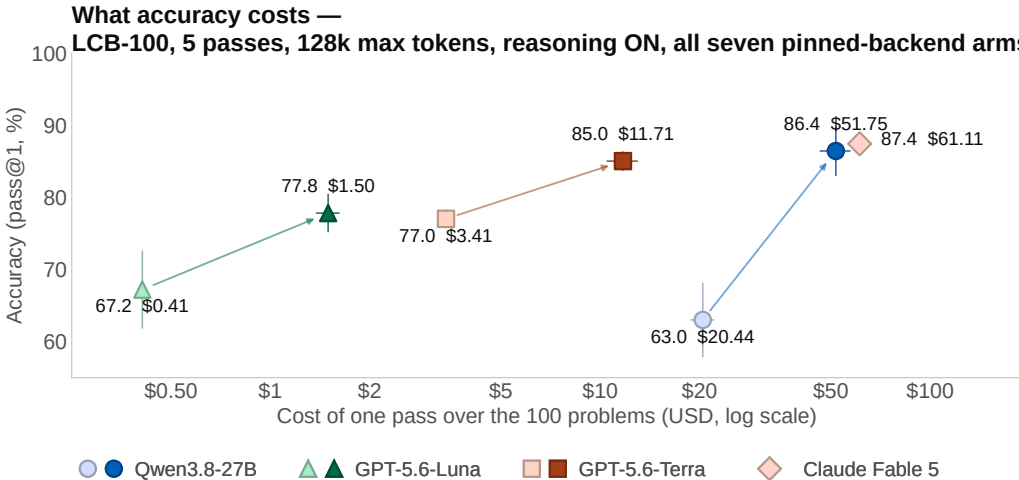


Figure 4: **Cost against accuracy.** One point per arm; an arrow runs from the single call to the manager of each model that has both. x is dollars for one pass over the 100 problems (log scale), y is pass@1; the bar through each point is the 95% CI across the 5 passes (t, df = 4). Light fill = single call, dark = with manager; marker shape is the model. Qwen3.8-27B’s single arm is the 128k cap-matched one on BOTH axes - score from the replay, output tokens capped at 128,000 per call to match. Priced as generated at 250k it would sit at \$29.08 rather than \$20.44. Retried and discarded attempts are counted: they were generated and would be billed. The cheapest arm is GPT-5.6-Luna, single call at \$0.41 a pass and the most accurate is Fable 5, single call at \$61.11 - a 149 $\times$ spread in price for +20.2 points.

Table 2: **Cap hits and empty solutions per arm.** Single (s) / manager (m), out of 500 problem-passes each. A *cap hit* is a graded generation that reached the token limit (`finish_reason=length`): the single arm’s one call, and for the manager its final answer call only — internally 108 of its 3,235 calls also hit the cap, 95 of them workers, which the loop absorbs. *No code* is a problem-pass that ended with nothing the extractor could grade.

Model	Cap hits s / m	No code s / m	of which refusals	Cap
GPT-5.6-Terra	0 / 0	0 / 0	0	128k ^b
GPT-5.6-Luna	0 / 0	0 / 0	0	128k ^b
Qwen3.8-27B	150 / 5	35 / 0	0	128k ^a
Claude Fable 5	3 / –	9 / –	6	128k ^b

^a Every count in this table is at a 128k cap, so the rows are like-for-like. Qwen3.8-27B’s manager arm ran natively at 128k; its single arm was run at 250k and is counted here cap-matched back to 128k. The 250k counts are given in Table 3.

^b A *hard* ceiling rather than a chosen setting: the GPT-5.6 family and Fable 5 both cap a synchronous API response at 128k output tokens (Appendix B), so for these three arms a truncation cannot be relieved by raising the cap. Qwen3.8-27B, served locally, has no such ceiling — which is why its single arm could be run at 250k at all.

Table 3: **Qwen3.8-27B’s single arm read at both caps.** Cap-matched to 128k, and as generated at 250k.

Qwen3.8-27B single	pass@1	Cut off by the cap	No code at all
128k (cap-matched, App. B)	63.0 ± 4.1	150 / 500 (30.0 %)	35 / 500 (7.0 %)
250k (as generated)	65.6 ± 4.6	124 / 500 (24.8 %)	21 / 500 (4.2 %)

Table 4: **Each single arm re-scored** over only the problem-passes where it emitted code.

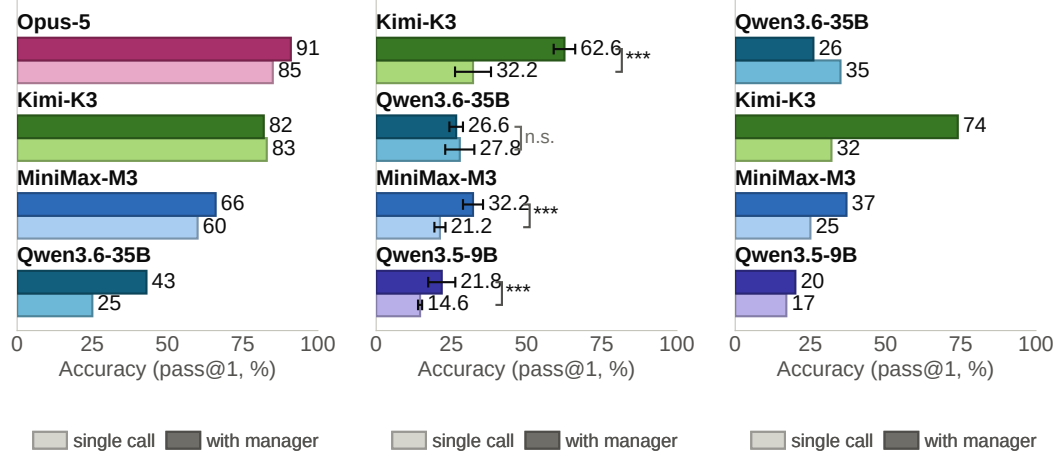
Model (single arm)	As scored	Emitted code	Restricted to those
GPT-5.6-Terra	77.0	500/500	77.0
GPT-5.6-Luna	67.2	500/500	67.2
Qwen3.8-27B	63.0	465/500	67.7 (+4.7)
Claude Fable 5	87.4	491/500	89.0 (+1.6)

Table 5: **LCB-100 pass@1 (%)**, **single** → **manager**. For the OpenRouter-served models on the original scaffold. “–” = not run; Qwen3.5-9B reasoning-on returns reasoning-only replies and is unusable. The seven pinned-backend, five-pass arms are reported separately in Section 5 and are not pooled here, because both the serving path and the scaffold version differ (Section 4).

Model	Params	128k · ON (1 pass)	128k · OFF (1 pass)	16k · OFF (×5)
Opus-5	n/a	85 → 91 (+6)	–	–
Kimi-K3	~2.8T	83 → 82 (–1)	32 → 74 (+42)	32.2 → 62.6 (+30.4)
MiniMax-M3	428B	60 → 66 (+6)	25 → 37 (+12)	21.2 → 32.2 (+11.0)
Qwen3.6-35B	35B	25 → 43 (+18)	35 → 26 (–9)	27.8 → 26.6 (–1.2)
Qwen3.5-9B	9B	<i>unusable</i>	17 → 20 (+3)	14.6 → 21.8 (+7.2)

Table 6: **One transcript per model where the single call failed and the manager arm solved the problem** (Section 6). LCB problem IDs as in release_v6; call counts are manager-arm model calls. Character counts are of the emitted program.

Model (condition)	Problem	How the single call failed	What the scaffold added
Qwen3.5-9B (128k, off)	abc385_d	Timed out: a naïve per-segment scan of a path simulation.	The brainstorm flagged in writing that the check is $O(N \cdot M) \approx 4 \times 10^{10}$ and prescribed a range-query structure <i>before</i> any code — the up-front planning a single pass skips (Section 6).
Qwen3.6-35B (on)	abc394_f	Cut off mid-reasoning at 32,768 tokens; returned nothing.	Solved in one worker cycle: the brainstorm fixed the structural insight (an “alkane” subtree needs a degree-4 center and ≥ 5 vertices); the worker used an iterative DFS to avoid a recursion-depth failure. The same 32k clamp was hit 115 times across this model’s manager calls, but with several attempts per problem one clamped call no longer costs the problem.
MiniMax-M3 (on)	3701	Printed nothing on <code>cdcd</code> : the lexicographic-reconstruction branch was broken.	Split the two coupled objectives across worker cycles — one worker on the forward cost DP over capped run-lengths, a separate worker on a suffix DP used purely for the lexicographic reconstruction.
Kimi-K3 (128k, off)	3687	Over-counted the longest unique-value path (9 and 3 where the answers were 6 and 2), never shrinking its window on a repeated value.	The brainstorm wrote the reduction into <code>notes.md</code> first — longest substring without repeating characters along each root-to-leaf path, window start via last-seen depths, $O(n)$ DFS — and only then did a worker implement it.
GPT-5.6-Terra (on)	3688	4,615 characters implementing a Segment-Tree-Beats-shaped structure indexed by candidate value, and wrong.	The plan named the difficulty in advance — the state must be compressed rather than maintaining all distinct values per index — and nine calls later the manager arm returned 1,860 characters over one standard segment tree. The contribution is <i>subtraction</i> . Manager-only win in four of five passes.
GPT-5.6-Luna (on)	abc397_e	Carried a <i>set</i> of unfinished path lengths per subtree across 4,092 characters, a general state it never closed.	The brainstorm established the bounding lemma (at most two unfinished paths at any vertex), after which a worker implemented the postorder scan directly in 2,166 characters over five calls. Manager-only win in two of five passes.
Qwen3.8-27B (on)	abc397_g	Spent all 250,000 tokens on edge-case self-interrogation and never emitted a program — the runaway-reasoning failure mode of Section 6 in pure form — though the reasoning had already reached the min-cut formulation.	Committed that same reduction to <code>plan.md</code> as a plan, then had a worker implement it: five calls, 3,548 characters, manager-only win in all five passes. The scaffold does not supply the insight; it forces it onto disk before the budget runs out.
Opus-5 (on)	abc388_e	The feasibility test over-counted, reporting 225 pairs where 220 is optimal.	Recorded the <i>justification</i> in <code>notes.md</code> first — an exchange argument, plus monotonicity of feasibility in K licensing a binary search — then had one worker implement it and a second harden it. A genuine correction, not truncation relief.



(a) 128k max tokens, reasoning on, one pass per condition. (b) 16k max tokens, reasoning off, five passes per condition. (c) 128k max tokens, reasoning off, one pass per condition.

Figure 5: Manager vs. single call, three conditions. Models run top to bottom by single-call score, best first, so a model’s row differs between panels; the top bar of each pair is the manager. Fill: light = single call (one call, no tools), dark = with manager; Δ = manager – single, in percentage points. The line through a bar is the 95% CI across its passes, where the condition ran more than one. A panel prints its two bar values and nothing else; the Δ and Tukey HSD group letter quoted below per model are what it leaves out (shared letter = not significantly different across the models of that condition, $\alpha = 0.05$). (a) 128k max tokens, reasoning on - effort:high for Kimi-K3 and Opus-5, a 20k reasoning budget requested for Qwen3.6-35B and MiniMax-M3 that providers often did not honor. One pass per condition, so no per-model p. Non-empty completions per 100 (single→manager): Opus-5 97→100, Kimi-K3 97→100, MiniMax-M3 92→93, Qwen3.6-35B 34→68. Empty or truncated output scores as a fail, so Δ partly tracks emit rate. Opus-5 Δ +6.0 (group ab), Kimi-K3 Δ -1.0 (group b), MiniMax-M3 Δ +6.0 (group ab), Qwen3.6-35B Δ +18.0 (group a). (b) 16k max tokens, reasoning off, five passes per condition; the line through each bar is the 95% CI across the five passes (t, df = 4). Each Δ is a paired sign-flip permutation test, unit = problem (n = 100), Holm-corrected across the 4 models; the bracket beside a pair grades its p: * p < .05, ** p < .01, *** p < .001, n.s. otherwise. Kimi-K3 Δ +30.4 (group a), Qwen3.6-35B Δ -1.2 (group c), MiniMax-M3 Δ +11.0 (group b), Qwen3.5-9B Δ +7.2 (group bc). (c) 128k max tokens, reasoning off (ESCALATION_OR_REASONING=none), one pass per condition, so no per-model p. Non-empty completions per 100 (single→manager): Qwen3.6-35B 85→95, Kimi-K3 96→100, MiniMax-M3 93→100, Qwen3.5-9B 79→92. Qwen3.6-35B Δ -9.0 (group c), Kimi-K3 Δ +42.0 (group a), MiniMax-M3 Δ +12.0 (group b), Qwen3.5-9B Δ +3.0 (group bc).

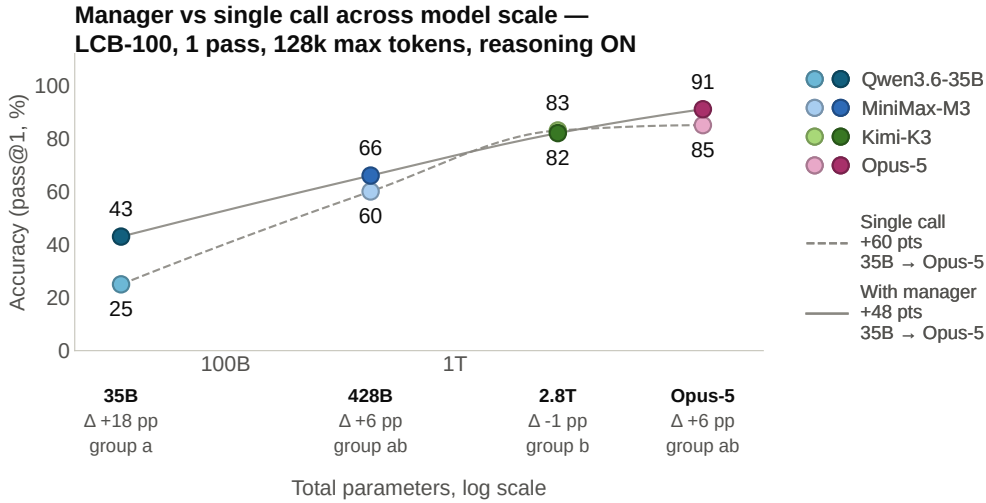


Figure 6: **Accuracy by model scale, reasoning on, 128k, one pass.** The same numbers as Figure 5a, against parameter count. Dashed = single call, solid = with manager; the block under each tick gives that model’s Δ and Tukey group, and the legend each arm’s gain from the smallest model to the largest.

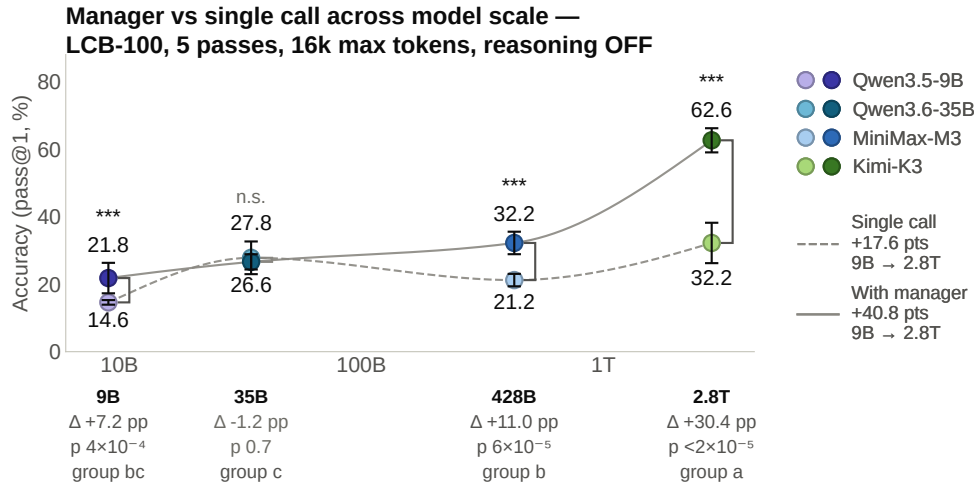


Figure 7: **Accuracy by model scale, reasoning off, 16k, five passes.** The same numbers as Figure 5b, against parameter count; the line through each mark is the 95% CI across the five passes. Dashed = single call, solid = with manager, and the bracket at a pair grades its Holm-corrected p; the block under each tick gives that model’s Δ , p and Tukey group, and the legend each arm’s gain from 9B to 2.8T.

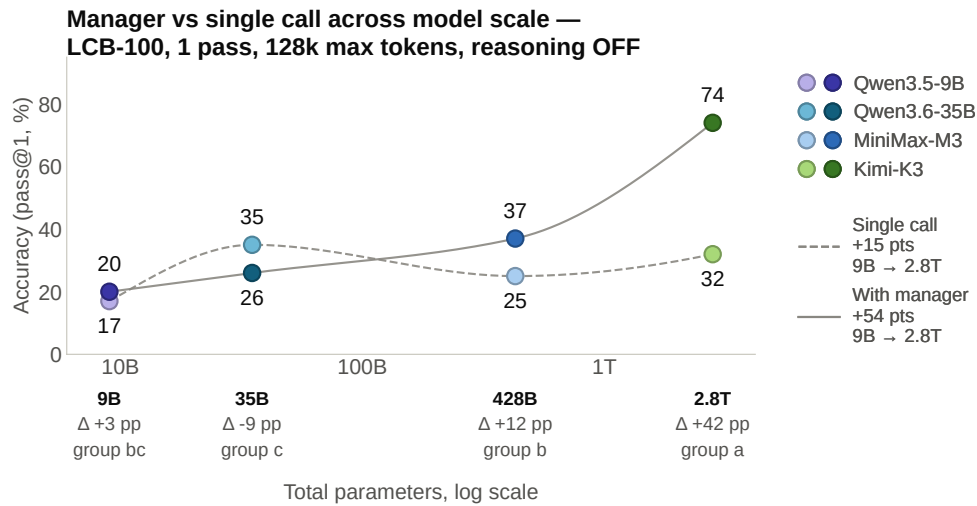


Figure 8: **Accuracy by model scale, reasoning off, 128k, one-pass control.** The same numbers as Figure 5c, against parameter count. Dashed = single call, solid = with manager; the block under each tick gives that model’s Δ and Tukey group, and the legend each arm’s gain from the smallest model to the largest.